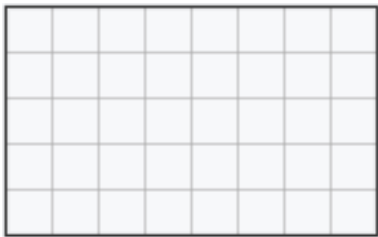
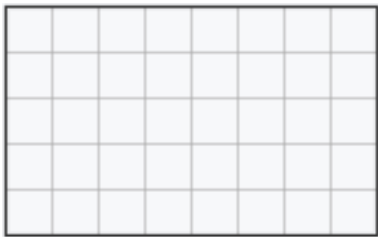


Mini Lesson Sample Script

Materials: Centimeter grid paper, personal white board

1. Find the area of two equal rectangles and add to find the total area.

Here are 2 equal rectangles. Label the side lengths. Write an equation to find the area of each rectangle. Then write an equation to show the total area.



Answer: Side lengths: 5 units and 8 units for each rectangle; $5 \times 8 = 40$ sq units; $5 \times 8 = 40$ sq units; $40 + 40 = 80$ sq units

T: Take a look at these two rectangles. Count the side lengths and label them on the first rectangle.

S: This side is 5 units, and this side is 8 units.

T: Good. How can you find the area of this rectangle?

S: I can multiply the side lengths. 5 times 8 is 40 square units.

T: Write the equation. Now look at the other rectangle. What's its area?

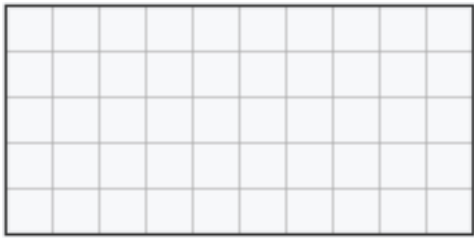
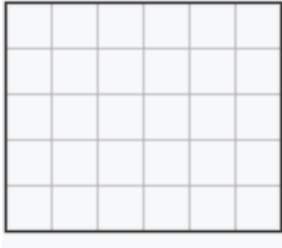
S: It's also 40 square units, because both rectangles are the same size.

T: Right, they're equal rectangles. Now write an addition equation to find the total area of both rectangles.

S: 40 plus 40 equals 80 square units.

	T: So the total area of the two rectangles is 80 square units.
2. Combine two equal rectangles into a longer rectangle and find its area.	
Push the two equal rectangles from Problem 1 together along the matching side to make one longer rectangle. Draw the new rectangle and label the side lengths. What is the area of the longer rectangle? <i>Answer: Side lengths: 5 units and 16 units; Area = 80 sq units</i>	T: Now imagine sliding these two rectangles next to each other so the matching sides of 5 units line up. Draw the new, longer rectangle. What are the side lengths? S: One side is still 5 units. The longer side is 8 plus 8, that's 16 units. T: Before you compute, what do you think the area of this longer rectangle is? S: I think it's 80 square units, because that's the total we got in Problem 1. T: Why does that work? S: We didn't add any new squares. We just put the two rectangles together. So the area has to be the same. T: Exactly. When we push the rectangles together, we're just rearranging the same square units. The longer rectangle has an area of 80 square units, the same as 40 plus 40.
3. Find the area of a longer rectangle by adding the areas of two unequal rectangles joined together.	

Here are 2 rectangles that share a side length of 5 units. One rectangle is 5 units by 6 units, and the other is 5 units by 10 units. Find the area of each rectangle. Then push them together along the matching side to make one longer rectangle. What is the total area of the longer rectangle?



Answer: $5 \times 6 = 30$ sq units; $5 \times 10 = 50$ sq units; $30 + 50 = 80$ sq units

T: Label the side lengths on each rectangle, then find the area of each.

S: The first one is 5 by 6, so the area is 5 times 6, that's 30 square units. The second one is 5 by 10, so the area is 50 square units.

T: Now draw the longer rectangle made by joining these two along the matching side of 5 units. What are the side lengths?

S: One side is still 5 units, and the other side is 6 plus 10, that's 16 units.

T: Find the total area.

S: 30 plus 50 is 80 square units.

T: How do you know the area of the longer rectangle is 80 square units without multiplying 5 times 16?

S: Because the longer rectangle is just the two smaller rectangles joined together. So I can add their areas to get the total.

T: Adding the areas of the two smaller rectangles gives you the total area of the longer rectangle, even when the two rectangles are not equal.